

SCR2043 OPERATING SYSTEMS

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Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. Download necessary source codes:

Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. Compile the source files:

Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. Execute the dummy processes:

Run all the dummy processes

```
./mainprocess &
```

Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
<code>ps -e</code>
Output

```

ps -e
  PID TTY          TIME CMD
    1 ?           00:00:05 systemd
    2 ?           00:00:00 kthreadd
    3 ?           00:00:00 pool_workqueue_release
    4 ?           00:00:00 kworker/R-rcu_g
    5 ?           00:00:00 kworker/R-rcu_p
    6 ?           00:00:00 kworker/R-slub_
    7 ?           00:00:00 kworker/R-netns
    8 ?           00:00:00 kworker/0:0-cgroup_destroy
   10 ?           00:00:00 kworker/0:0H-kblockd
   11 ?           00:00:00 kworker/u4:0-ext4-rsv-conversion
   12 ?           00:00:00 kworker/R-mm_pe
   13 ?           00:00:00 rcu_tasks_kthread
   14 ?           00:00:00 rcu_tasks_rude_kthread
   15 ?           00:00:00 rcu_tasks_trace_kthread
   16 ?           00:00:00 ksoftirqd/0
   17 ?           00:00:00 rcu_preempt
   18 ?           00:00:00 migration/0
   19 ?           00:00:00 idle_inject/0
   20 ?           00:00:00 cpuhp/0
   21 ?           00:00:00 cpuhp/1
   22 ?           00:00:00 idle_inject/1
   23 ?           00:00:00 migration/1
   24 ?           00:00:00 ksoftirqd/1
   25 ?           00:00:00 kworker/1:0-cgroup_destroy
   27 ?           00:00:00 kworker/u5:0-events_power_efficient
   29 ?           00:00:00 kdevtmpfs
   30 ?           00:00:00 kworker/R-inet_
   31 ?           00:00:00 kworker/u5:1-events_power_efficient
   32 ?           00:00:00 kauditd
   33 ?           00:00:00 khungtaskd
   34 ?           00:00:00 oom_reaper
   36 ?           00:00:00 kworker/R-write
   37 ?           00:00:00 kcompactd0
   38 ?           00:00:00 ksm
   40 ?           00:00:00 khugepaged
   41 ?           00:00:00 kworker/R-kinte
   42 ?           00:00:00 kworker/R-kbloc
   43 ?           00:00:00 kworker/R-blkcg
   44 ?           00:00:00 irq/9-acpi
   45 ?           00:00:00 kworker/R-tpm_d
   46 ?           00:00:00 kworker/R-ata_s
   47 ?           00:00:00 kworker/R-md
   48 ?           00:00:00 kworker/R-md_bi
   49 ?           00:00:00 kworker/R-edac-
   50 ?           00:00:00 kworker/R-devfr
   51 ?           00:00:00 watchdogd
   53 ?           00:00:00 kworker/1:1H-kblockd
   54 ?           00:00:00 kswapd0
   55 ?           00:00:00 ecryptfs-kthread
   56 ?           00:00:00 kworker/R-kthro
   57 ?           00:00:00 kworker/R-acpi_
   58 ?           00:00:00 scsi_eh_0
   59 ?           00:00:00 kworker/R-scsi_
   60 ?           00:00:00 scsi_eh_1
   61 ?           00:00:00 kworker/R-scsi_
   64 ?           00:00:00 kworker/R-mld
   65 ?           00:00:00 kworker/R-ipv6_
   72 ?           00:00:00 kworker/R-kstrp

```

```

74 ?      00:00:00 kworker/u7:0
75 ?      00:00:00 kworker/u8:0
76 ?      00:00:00 kworker/u9:0
78 ?      00:00:00 kworker/u6:2-events_power_efficient
82 ?      00:00:00 kworker/R-crypt
92 ?      00:00:00 kworker/R-charg
131 ?     00:00:00 kworker/u4:1
137 ?     00:00:07 kworker/0:3-events
154 ?     00:00:00 kworker/R-mpt_p
155 ?     00:00:00 kworker/R-mpt/0
156 ?     00:00:00 scsi_eh_2
157 ?     00:00:00 kworker/R-scsi_
191 ?     00:00:00 kworker/R-raid5
214 ?     00:00:00 kworker/0:2H-kblockd
229 ?     00:00:00 jbd2/sda1-8
230 ?     00:00:00 kworker/R-ext4-
297 ?     00:00:00 psimon
302 ?     00:00:00 systemd-journal
322 ?     00:00:00 kworker/R-kmpat
323 ?     00:00:00 kworker/R-kmpat
358 ?     00:00:00 multipathd
365 ?     00:00:00 systemd-udev
371 ?     00:00:00 psimon
442 ?     00:00:00 irq/18-vmwgfx
445 ?     00:00:00 kworker/R-ttm
482 ?     00:00:00 jbd2/sda16-8
483 ?     00:00:00 kworker/R-ext4-
513 ?     00:00:00 systemd-network
524 ?     00:00:00 systemd-resolve
527 ?     00:00:00 systemd-timesyn

547 ?     00:00:02 kworker/1:3-ata_sff
639 ?     00:00:00 dbus-daemon
645 ?     00:00:00 polkitd
656 ?     00:00:00 systemd-logind
658 ?     00:00:00 udiskd
684 ?     00:00:00 rsyslogd
716 ?     00:00:00 unattended-upgr
729 ?     00:00:00 cron
756 ?     00:00:00 ModemManager
788 tty1   00:00:00 agetty
891 ?     00:00:00 kworker/1:2H
904 ?     00:00:00 kworker/u5:4-events_power_efficient
925 ?     00:00:00 sshd
926 ?     00:00:00 sshd
928 ?     00:00:02 sshd
929 pts/0   00:00:00 bash
969 ?     00:00:00 kworker/u6:0-events_power_efficient
984 ?     00:00:03 fwupd
1019 ?    00:00:00 kworker/1:1
1137 ?    00:00:00 kworker/u5:2-events_power_efficient
1142 ?    00:00:00 kworker/0:1
1143 ?    00:00:00 kworker/0:2-events
1150 pts/0   00:00:00 mainprocess
1151 pts/0   00:00:00 mainprocess
1152 pts/0   00:00:00 mainprocess
1153 pts/0   00:00:00 subprocess2
1154 pts/0   00:00:00 subprocess2
1155 pts/0   00:00:00 subprocess2
1156 pts/0   00:00:00 subprocess1
1157 pts/0   00:00:00 subprocess1

```

1158	?	00:00:00	kworker/u6:1-events_unbound
1159	pts/0	00:00:00	mainprocess
1160	pts/0	00:00:00	mainprocess
1161	pts/0	00:00:00	mainprocess
1162	pts/0	00:00:00	subprocess1
1163	pts/0	00:00:00	subprocess1
1164	pts/0	00:00:00	subprocess2
1165	pts/0	00:00:00	subprocess2
1166	pts/0	00:00:00	subprocess2
1167	?	00:00:00	kworker/u6:3
1168	pts/0	00:00:00	ps

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command							
<code>ps -ef</code>							
Output							
UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	03:18	?	00:00:05	/sbin/init
root	2	0	0	03:18	?	00:00:00	[kthreadd]
root	3	2	0	03:18	?	00:00:00	[pool_workqueue_release]
root	4	2	0	03:18	?	00:00:00	[kworker/R-rcu_g]
root	5	2	0	03:18	?	00:00:00	[kworker/R-rcu_p]
root	6	2	0	03:18	?	00:00:00	[kworker/R-slub_]
root	7	2	0	03:18	?	00:00:00	[kworker/R-netns]
root	10	2	0	03:18	?	00:00:00	[kworker/0:0H-kblockd]
root	11	2	0	03:18	?	00:00:00	[kworker/u4:0-ext4-rsv-conversion]
root	12	2	0	03:18	?	00:00:00	[kworker/R-mm_pe]
root	13	2	0	03:18	?	00:00:00	[rcu_tasks_kthread]
root	14	2	0	03:18	?	00:00:00	[rcu_tasks_rude_kthread]
root	15	2	0	03:18	?	00:00:00	[rcu_tasks_trace_kthread]
root	16	2	0	03:18	?	00:00:00	[ksoftirqd/0]
root	17	2	0	03:18	?	00:00:00	[rcu_preempt]
root	18	2	0	03:18	?	00:00:00	[migration/0]
root	19	2	0	03:18	?	00:00:00	[idle_inject/0]
root	20	2	0	03:18	?	00:00:00	[cpuhp/0]
root	21	2	0	03:18	?	00:00:00	[cpuhp/1]
root	22	2	0	03:18	?	00:00:00	[idle_inject/1]
root	23	2	0	03:18	?	00:00:00	[migration/1]
root	24	2	0	03:18	?	00:00:00	[ksoftirqd/1]
root	25	2	0	03:18	?	00:00:00	[kworker/1:0-cgroup_destroy]
root	27	2	0	03:18	?	00:00:00	[kworker/u5:0-flush-8:0]
root	29	2	0	03:18	?	00:00:00	[kdevtmpfs]
root	30	2	0	03:18	?	00:00:00	[kworker/R-inet_]
root	31	2	0	03:18	?	00:00:00	[kworker/u5:1-events_unbound]
root	32	2	0	03:18	?	00:00:00	[kauditd]
root	33	2	0	03:18	?	00:00:00	[khungtaskd]
root	34	2	0	03:18	?	00:00:00	[oom_reaper]
root	36	2	0	03:18	?	00:00:00	[kworker/R-write]
root	37	2	0	03:18	?	00:00:00	[kcompactd0]
root	38	2	0	03:18	?	00:00:00	[ksmd]
root	40	2	0	03:18	?	00:00:00	[khugepaged]
root	41	2	0	03:18	?	00:00:00	[kworker/R-kinte]

root	42	2	0	03:18	?	00:00:00	[kworker/R-kbloc]
root	43	2	0	03:18	?	00:00:00	[kworker/R-blkcg]
root	44	2	0	03:18	?	00:00:00	[irq/9-acpi]
root	45	2	0	03:18	?	00:00:00	[kworker/R-tpm_d]
root	46	2	0	03:18	?	00:00:00	[kworker/R-ata_s]
root	47	2	0	03:18	?	00:00:00	[kworker/R-md]
root	48	2	0	03:18	?	00:00:00	[kworker/R-md_bi]
root	49	2	0	03:18	?	00:00:00	[kworker/R-edac-]
root	50	2	0	03:18	?	00:00:00	[kworker/R-devfr]
root	51	2	0	03:18	?	00:00:00	[watchdogd]
root	53	2	0	03:18	?	00:00:00	[kworker/1:1H-kblockd]
root	54	2	0	03:18	?	00:00:00	[kswapd0]
root	55	2	0	03:18	?	00:00:00	[ecryptfs-kthread]
root	56	2	0	03:18	?	00:00:00	[kworker/R-kthro]
root	57	2	0	03:18	?	00:00:00	[kworker/R-acpi_]
root	58	2	0	03:18	?	00:00:00	[scsi_eh_0]
root	59	2	0	03:18	?	00:00:00	[kworker/R-scsi_]
root	60	2	0	03:18	?	00:00:00	[scsi_eh_1]
root	61	2	0	03:18	?	00:00:00	[kworker/R-scsi_]
root	64	2	0	03:18	?	00:00:00	[kworker/R-mld]
root	65	2	0	03:18	?	00:00:00	[kworker/R-ipv6_]
root	72	2	0	03:18	?	00:00:00	[kworker/R-kstrp]
root	74	2	0	03:18	?	00:00:00	[kworker/u7:0]
root	75	2	0	03:18	?	00:00:00	[kworker/u8:0]
root	76	2	0	03:18	?	00:00:00	[kworker/u9:0]
root	82	2	0	03:18	?	00:00:00	[kworker/R-crypt]
root	92	2	0	03:18	?	00:00:00	[kworker/R-charg]
root	131	2	0	03:18	?	00:00:00	[kworker/u4:1]
root	137	2	0	03:18	?	00:00:09	[kworker/0:3-events]
root	154	2	0	03:18	?	00:00:00	[kworker/R-mpt_p]
root	155	2	0	03:18	?	00:00:00	[kworker/R-mpt/0]
root	156	2	0	03:18	?	00:00:00	[scsi_eh_2]
root	157	2	0	03:18	?	00:00:00	[kworker/R-scsi_]
root	191	2	0	03:18	?	00:00:00	[kworker/R-raid5]
root	214	2	0	03:18	?	00:00:00	[kworker/0:2H-kblockd]
root	229	2	0	03:18	?	00:00:00	[jbd2/sda1-8]
root	230	2	0	03:18	?	00:00:00	[kworker/R-ext4-]
root	297	2	0	03:18	?	00:00:00	[psimon]
root	302	1	0	03:18	?	00:00:00	/usr/lib/systemd/systemd-journald

```

root      322      2 0 03:18 ?      00:00:00 [kworker/R-kmpat]
root      323      2 0 03:18 ?      00:00:00 [kworker/R-kmpat]
root      358      1 0 03:18 ?      00:00:00 /sbin/multipathd -d -s
root      365      1 0 03:18 ?      00:00:00 /usr/lib/systemd/systemd-udev
root      371      2 0 03:18 ?      00:00:00 [psimon]
root      442      2 0 03:18 ?      00:00:00 [irq/18-vmwgfx]
root      445      2 0 03:18 ?      00:00:00 [kworker/R-ttm]
root      482      2 0 03:18 ?      00:00:00 [jbd2/sda16-8]
root      483      2 0 03:18 ?      00:00:00 [kworker/R-ext4-]
systemd+  513      1 0 03:18 ?      00:00:00 /usr/lib/systemd/systemd-networkd
systemd+  524      1 0 03:18 ?      00:00:00 /usr/lib/systemd/systemd-resolved
systemd+  527      1 0 03:18 ?      00:00:00 /usr/lib/systemd/systemd-timesyncd
root      547      2 0 03:18 ?      00:00:03 [kworker/1:3-events]
message+  639      1 0 03:18 ?      00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd+
polkitd   645      1 0 03:18 ?      00:00:00 /usr/lib/polkit-1/polkitd --no-debug
root      656      1 0 03:18 ?      00:00:00 /usr/lib/systemd/systemd-logind
root      658      1 0 03:18 ?      00:00:00 /usr/libexec/udisks2/udisksd
syslog    684      1 0 03:18 ?      00:00:00 /usr/sbin/rsyslogd -n -iNONE
root      716      1 0 03:18 ?      00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-s
root      729      1 0 03:18 ?      00:00:00 /usr/sbin/cron -f -P
root      756      1 0 03:18 ?      00:00:00 /usr/sbin/ModemManager
root      788      1 0 03:18 tty1    00:00:00 /sbin/agetty -o -p -- \u --noclear - linux
root      891      2 0 03:19 ?      00:00:00 [kworker/1:2H]
root      904      2 0 03:19 ?      00:00:00 [kworker/u5:4-events_power_efficient]
root      925      1 0 03:20 ?      00:00:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root      926      925 0 03:20 ?      00:00:00 sshd: chew [priv]
chew      926      926 0 03:20 ?      00:00:02 sshd: chew@pts/0
chew      929      928 0 03:20 pts/0    00:00:00 -bash
root      984      1 0 03:25 ?      00:00:03 /usr/libexec/fwupd/fwupd
root      1137     2 0 03:26 ?      00:00:00 [kworker/u5:2-events_power_efficient]
root      1143     2 0 03:30 ?      00:00:00 [kworker/0:2-events]
chew      1150     929 0 03:31 pts/0    00:00:00 ./mainprocess
chew      1151     1150 0 03:31 pts/0    00:00:00 ./mainprocess
chew      1152     1150 0 03:31 pts/0    00:00:00 ./mainprocess
chew      1153     1152 0 03:31 pts/0    00:00:00 ./subprocess2
chew      1154     1152 0 03:31 pts/0    00:00:00 ./subprocess2
chew      1155     1152 0 03:31 pts/0    00:00:00 ./subprocess2
chew      1156     1151 0 03:31 pts/0    00:00:00 ./subprocess1
chew      1157     1151 0 03:31 pts/0    00:00:00 ./subprocess1

root      1158     2 0 03:31 ?      00:00:00 [kworker/u6:1-events_unbound]
chew      1159     929 0 03:31 pts/0    00:00:00 ./mainprocess
chew      1160     1159 0 03:31 pts/0    00:00:00 ./mainprocess
chew      1161     1159 0 03:31 pts/0    00:00:00 ./mainprocess
chew      1162     1160 0 03:31 pts/0    00:00:00 ./subprocess1
chew      1163     1160 0 03:31 pts/0    00:00:00 ./subprocess1
chew      1164     1161 0 03:31 pts/0    00:00:00 ./subprocess2
chew      1165     1161 0 03:31 pts/0    00:00:00 ./subprocess2
chew      1166     1161 0 03:31 pts/0    00:00:00 ./subprocess2
root      1167     2 0 03:32 ?      00:00:00 [kworker/u6:3-events_unbound]
root      1176     2 0 03:37 ?      00:00:00 [kworker/u6:0-events_unbound]
chew      1177     929 99 03:37 pts/0    00:00:00 ps -ef

```


Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command									
<code>ps -ef grep subprocess</code>									
Output									
chew	1153	1152	0	03:31	pts/0	00:00:00	./subprocess2		
chew	1154	1152	0	03:31	pts/0	00:00:00	./subprocess2		
chew	1155	1152	0	03:31	pts/0	00:00:00	./subprocess2		
chew	1156	1151	0	03:31	pts/0	00:00:00	./subprocess1		
chew	1157	1151	0	03:31	pts/0	00:00:00	./subprocess1		
chew	1162	1160	0	03:31	pts/0	00:00:00	./subprocess1		
chew	1163	1160	0	03:31	pts/0	00:00:00	./subprocess1		
chew	1164	1161	0	03:31	pts/0	00:00:00	./subprocess2		
chew	1165	1161	0	03:31	pts/0	00:00:00	./subprocess2		
chew	1166	1161	0	03:31	pts/0	00:00:00	./subprocess2		
chew	1180	929	50	03:39	pts/0	00:00:00	grep --color=auto subprocess		

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command				
<code>ps -eo pid,user,pcpu,pmem,cmd</code>				
Output				
PID	USER	%CPU	%MEM	CMD
1	root	0.4	1.3	/sbin/init
2	root	0.0	0.0	[kthreadd]
3	root	0.0	0.0	[pool_workqueue_release]
4	root	0.0	0.0	[kworker/R-rcu_g]
5	root	0.0	0.0	[kworker/R-rcu_p]
6	root	0.0	0.0	[kworker/R-slub_]
7	root	0.0	0.0	[kworker/R-netns]
10	root	0.0	0.0	[kworker/0:0H-kblockd]
11	root	0.0	0.0	[kworker/u4:0-ext4-rsv-conversion]
12	root	0.0	0.0	[kworker/R-mm_pe]
13	root	0.0	0.0	[rcu_tasks_kthread]
14	root	0.0	0.0	[rcu_tasks_rude_kthread]
15	root	0.0	0.0	[rcu_tasks_trace_kthread]
16	root	0.0	0.0	[ksoftirqd/0]
17	root	0.0	0.0	[rcu_preempt]
18	root	0.0	0.0	[migration/0]
19	root	0.0	0.0	[idle_inject/0]
20	root	0.0	0.0	[cpuhp/0]
21	root	0.0	0.0	[cpuhp/1]
22	root	0.0	0.0	[idle_inject/1]
23	root	0.0	0.0	[migration/1]
24	root	0.0	0.0	[ksoftirqd/1]
25	root	0.0	0.0	[kworker/1:0-events]
27	root	0.0	0.0	[kworker/u5:0-events_power_efficient]
29	root	0.0	0.0	[kdevtmpfs]
30	root	0.0	0.0	[kworker/R-inet_]
31	root	0.0	0.0	[kworker/u5:1-events_power_efficient]
32	root	0.0	0.0	[kauditd]
33	root	0.0	0.0	[khungtaskd]
34	root	0.0	0.0	[oom_reaper]
36	root	0.0	0.0	[kworker/R-write]
37	root	0.0	0.0	[kcompactd0]
38	root	0.0	0.0	[ksmd]
40	root	0.0	0.0	[khugepaged]

41	root	0.0	0.0	[kworker/R-kinte]
42	root	0.0	0.0	[kworker/R-kbloc]
43	root	0.0	0.0	[kworker/R-blkcg]
44	root	0.0	0.0	[irq/9-acpi]
45	root	0.0	0.0	[kworker/R-tpm_d]
46	root	0.0	0.0	[kworker/R-ata_s]
47	root	0.0	0.0	[kworker/R-md]
48	root	0.0	0.0	[kworker/R-md_bi]
49	root	0.0	0.0	[kworker/R-edac-]
50	root	0.0	0.0	[kworker/R-devfr]
51	root	0.0	0.0	[watchdogd]
53	root	0.0	0.0	[kworker/1:1H-kblockd]
54	root	0.0	0.0	[kswapd0]
55	root	0.0	0.0	[ecryptfs-kthread]
56	root	0.0	0.0	[kworker/R-kthro]
57	root	0.0	0.0	[kworker/R-acpi_]
58	root	0.0	0.0	[scsi_eh_0]
59	root	0.0	0.0	[kworker/R-scsi_]
60	root	0.0	0.0	[scsi_eh_1]
61	root	0.0	0.0	[kworker/R-scsi_]
64	root	0.0	0.0	[kworker/R-mld]
65	root	0.0	0.0	[kworker/R-ipv6_]
72	root	0.0	0.0	[kworker/R-kstrp]
74	root	0.0	0.0	[kworker/u7:0]
75	root	0.0	0.0	[kworker/u8:0]
76	root	0.0	0.0	[kworker/u9:0]
82	root	0.0	0.0	[kworker/R-crypt]
92	root	0.0	0.0	[kworker/R-charg]
131	root	0.0	0.0	[kworker/u4:1]
137	root	0.8	0.0	[kworker/0:3-mpt_poll_0]
154	root	0.0	0.0	[kworker/R-mpt_p]
155	root	0.0	0.0	[kworker/R-mpt/0]
156	root	0.0	0.0	[scsi_eh_2]
157	root	0.0	0.0	[kworker/R-scsi_]
191	root	0.0	0.0	[kworker/R-raid5]
214	root	0.0	0.0	[kworker/0:2H-kblockd]
229	root	0.0	0.0	[jbd2/sda1-8]
230	root	0.0	0.0	[kworker/R-ext4-]
297	root	0.0	0.0	[psimon]

302	root	0.0	1.1	/usr/lib/systemd/systemd-journald
322	root	0.0	0.0	[kworker/R-kmpat]
323	root	0.0	0.0	[kworker/R-kmpat]
358	root	0.0	2.7	/sbin/multipathd -d -s
365	root	0.0	0.7	/usr/lib/systemd/systemd-udev
371	root	0.0	0.0	[psimon]
442	root	0.0	0.0	[irq/18-vmwgfx]
445	root	0.0	0.0	[kworker/R-ttm]
482	root	0.0	0.0	[jbd2/sda16-8]
483	root	0.0	0.0	[kworker/R-ext4-]
513	systemd+	0.0	0.9	/usr/lib/systemd/systemd-networkd
524	systemd+	0.0	1.2	/usr/lib/systemd/systemd-resolved
527	systemd+	0.0	0.7	/usr/lib/systemd/systemd-timesyncd
547	root	0.2	0.0	[kworker/1:3-events]
639	message+	0.0	0.5	@dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
645	polkitd	0.0	0.8	/usr/lib/polkit-1/polkitd --no-debug
656	root	0.0	0.8	/usr/lib/systemd/systemd-logind
658	root	0.0	1.3	/usr/libexec/udisks2/udisksd
684	syslog	0.0	0.6	/usr/sbin/rsyslogd -n -iNONE
716	root	0.0	2.3	/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
729	root	0.0	0.2	/usr/sbin/cron -f -P
756	root	0.0	1.3	/usr/sbin/ModemManager
788	root	0.0	0.1	/sbin/agetty -o -p -- \u --noclear - linux
891	root	0.0	0.0	[kworker/1:2H]
904	root	0.0	0.0	[kworker/u5:4-events_unbound]
925	root	0.0	0.8	sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
926	root	0.0	0.5	sshd: chew [priv]
928	chew	0.2	0.7	sshd: chew@pts/0
929	chew	0.0	0.4	-bash
984	root	0.3	4.1	/usr/libexec/fwupd/fwupd
1137	root	0.0	0.0	[kworker/u5:2-events_power_efficient]
1143	root	0.0	0.0	[kworker/0:2-events]
1150	chew	0.0	0.1	./mainprocess
1151	chew	0.0	0.0	./mainprocess
1152	chew	0.0	0.0	./mainprocess
1153	chew	0.0	0.1	./subprocess2
1154	chew	0.0	0.1	./subprocess2
1155	chew	0.0	0.1	./subprocess2
1156	chew	0.0	0.1	./subprocess1
1157	chew	0.0	0.1	./subprocess1
1158	root	0.0	0.0	[kworker/u6:1-events_unbound]
1159	chew	0.0	0.1	./mainprocess
1160	chew	0.0	0.0	./mainprocess
1161	chew	0.0	0.0	./mainprocess
1162	chew	0.0	0.1	./subprocess1
1163	chew	0.0	0.1	./subprocess1
1164	chew	0.0	0.1	./subprocess2
1165	chew	0.0	0.1	./subprocess2
1166	chew	0.0	0.1	./subprocess2
1167	root	0.0	0.0	[kworker/u6:3-events_unbound]
1176	root	0.0	0.0	[kworker/u6:0-events_unbound]
1183	root	0.0	0.0	[kworker/1:1]
1184	chew	100	0.4	ps -eo pid,user,pcpu,pmem,cmd

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (pmem)?

Command
<code>ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem</code>

Output

```

PID USER      %CPU %MEM  CMD
984 root        0.2  4.1  /usr/libexec/fwupd/fwupd
358 root        0.0  2.7  /sbin/multipathd -d -s
716 root        0.0  2.3  /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
658 root        0.0  1.3  /usr/libexec/udisks2/udisksd
  1 root        0.2  1.3  /sbin/init
756 root        0.0  1.3  /usr/sbin/ModemManager
524 systemd+  0.0  1.2  /usr/lib/systemd/systemd-resolved
302 root        0.0  1.1  /usr/lib/systemd/systemd-journald
513 systemd+  0.0  0.9  /usr/lib/systemd/systemd-networkd
656 root        0.0  0.8  /usr/lib/systemd/systemd-logind
645 polkitd    0.0  0.8  /usr/lib/polkit-1/polkitd --no-debug
925 root        0.0  0.8  sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
527 systemd+  0.0  0.7  /usr/lib/systemd/systemd-timesyncd
365 root        0.0  0.7  /usr/lib/systemd/systemd-udev
928 chew       0.1  0.7  sshd: chew@pts/0
684 syslog    0.0  0.6  /usr/sbin/rsyslogd -n -iNONE
639 message+  0.0  0.5  @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
926 root        0.0  0.5  sshd: chew [priv]
929 chew       0.0  0.4  -bash
1194 chew      150  0.4  ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
729 root        0.0  0.2  /usr/sbin/cron -f -P
788 root        0.0  0.1  /sbin/agetty -o -p -- \u --noclear - linux
1150 chew      0.0  0.1  ./mainprocess
1153 chew      0.0  0.1  ./subprocess2
1154 chew      0.0  0.1  ./subprocess2
1155 chew      0.0  0.1  ./subprocess2
1156 chew      0.0  0.1  ./subprocess1
1157 chew      0.0  0.1  ./subprocess1
1159 chew      0.0  0.1  ./mainprocess
1162 chew      0.0  0.1  ./subprocess1
1163 chew      0.0  0.1  ./subprocess1
1164 chew      0.0  0.1  ./subprocess2

1165 chew      0.0  0.1  ./subprocess2
1166 chew      0.0  0.1  ./subprocess2
1151 chew      0.0  0.0  ./mainprocess
1152 chew      0.0  0.0  ./mainprocess
1160 chew      0.0  0.0  ./mainprocess
1161 chew      0.0  0.0  ./mainprocess
  2 root        0.0  0.0  [kthreadd]
  3 root        0.0  0.0  [pool_workqueue_release]
  4 root        0.0  0.0  [kworker/R-rcu_g]
  5 root        0.0  0.0  [kworker/R-rcu_p]
  6 root        0.0  0.0  [kworker/R-slub_]
  7 root        0.0  0.0  [kworker/R-netns]
 10 root        0.0  0.0  [kworker/0:0H-kblockd]
 11 root        0.0  0.0  [kworker/u4:0-ext4-rsv-conversion]
 12 root        0.0  0.0  [kworker/R-mm_pe]
 13 root        0.0  0.0  [rcu_tasks_kthread]
 14 root        0.0  0.0  [rcu_tasks_rude_kthread]
 15 root        0.0  0.0  [rcu_tasks_trace_kthread]
 16 root        0.0  0.0  [ksoftirqd/0]
 17 root        0.0  0.0  [rcu_preempt]
 18 root        0.0  0.0  [migration/0]
 19 root        0.0  0.0  [idle_inject/0]
 20 root        0.0  0.0  [cpuhp/0]
 21 root        0.0  0.0  [cpuhp/1]
 22 root        0.0  0.0  [idle_inject/1]
 23 root        0.0  0.0  [migration/1]
 24 root        0.0  0.0  [ksoftirqd/1]
 25 root        0.0  0.0  [kworker/1:0-events]
 27 root        0.0  0.0  [kworker/u5:0-events_power_efficient]
 29 root        0.0  0.0  [kdevtmpfs]
 30 root        0.0  0.0  [kworker/R-inet_]
 31 root        0.0  0.0  [kworker/u5:1-events_power_efficient]
 32 root        0.0  0.0  [kauditd]
 33 root        0.0  0.0  [khungtaskd]
 34 root        0.0  0.0  [oom_reaper]
 36 root        0.0  0.0  [kworker/R-write]
 37 root        0.0  0.0  [kcompactd0]
 38 root        0.0  0.0  [ksmd]
 40 root        0.0  0.0  [khugepaged]

```

```

41 root      0.0  0.0 [kworker/R-kinte]
42 root      0.0  0.0 [kworker/R-kbloc]
43 root      0.0  0.0 [kworker/R-blkcg]
44 root      0.0  0.0 [irq/9-acpi]
45 root      0.0  0.0 [kworker/R-tpm_d]
46 root      0.0  0.0 [kworker/R-ata_s]
47 root      0.0  0.0 [kworker/R-md]
48 root      0.0  0.0 [kworker/R-md_bi]
49 root      0.0  0.0 [kworker/R-edac-]
50 root      0.0  0.0 [kworker/R-devfr]
51 root      0.0  0.0 [watchdogd]
53 root      0.0  0.0 [kworker/1:1H-kblockd]
54 root      0.0  0.0 [kswapd0]
55 root      0.0  0.0 [ecryptfs-kthread]
56 root      0.0  0.0 [kworker/R-kthro]
57 root      0.0  0.0 [kworker/R-acpi_]
58 root      0.0  0.0 [scsi_eh_0]
59 root      0.0  0.0 [kworker/R-scsi_]
60 root      0.0  0.0 [scsi_eh_1]
61 root      0.0  0.0 [kworker/R-scsi_]
64 root      0.0  0.0 [kworker/R-mld]
65 root      0.0  0.0 [kworker/R-ipv6_]
72 root      0.0  0.0 [kworker/R-kstrp]
74 root      0.0  0.0 [kworker/u7:0]
75 root      0.0  0.0 [kworker/u8:0]
76 root      0.0  0.0 [kworker/u9:0]
82 root      0.0  0.0 [kworker/R-crypt]
92 root      0.0  0.0 [kworker/R-charg]
131 root     0.0  0.0 [kworker/u4:1]
137 root     0.7  0.0 [kworker/0:3-events]
154 root     0.0  0.0 [kworker/R-mpt_p]
155 root     0.0  0.0 [kworker/R-mpt/0]
156 root     0.0  0.0 [scsi_eh_2]
157 root     0.0  0.0 [kworker/R-scsi_]
191 root     0.0  0.0 [kworker/R-raid5]
214 root     0.0  0.0 [kworker/0:2H-kblockd]
229 root     0.0  0.0 [jbd2/sda1-8]
230 root     0.0  0.0 [kworker/R-ext4-]
297 root     0.0  0.0 [psimon]

322 root     0.0  0.0 [kworker/R-kmpat]
323 root     0.0  0.0 [kworker/R-kmpat]
371 root     0.0  0.0 [psimon]
442 root     0.0  0.0 [irq/18-vmwgfx]
445 root     0.0  0.0 [kworker/R-ttm]
482 root     0.0  0.0 [jbd2/sda16-8]
483 root     0.0  0.0 [kworker/R-ext4-]
547 root     0.1  0.0 [kworker/1:3-cgroup_destroy]
891 root     0.0  0.0 [kworker/1:2H]
904 root     0.0  0.0 [kworker/u5:4-events_unbound]
1137 root    0.0  0.0 [kworker/u5:2-events_unbound]
1143 root    0.0  0.0 [kworker/0:2-events]
1158 root    0.0  0.0 [kworker/u6:1-flush-8:0]
1167 root    0.0  0.0 [kworker/u6:3-events_unbound]
1176 root    0.0  0.0 [kworker/u6:0-events_power_efficient]
1189 root    0.0  0.0 [kworker/u5:3-events_unbound]
1193 root    0.0  0.0 [kworker/1:1]

```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command	
<pre>ps -ejH grep -E "mainprocess subprocess1 subprocess2"</pre>	
Output	
<pre>chew@secr2043:~\$ ps -ejH grep -E "mainprocess subprocess1 subprocess2" 1150 1150 929 pts/0 00:00:00 mainprocess 1151 1150 929 pts/0 00:00:00 mainprocess 1156 1150 929 pts/0 00:00:00 subprocess1 1157 1150 929 pts/0 00:00:00 subprocess1 1152 1150 929 pts/0 00:00:00 mainprocess 1153 1150 929 pts/0 00:00:00 subprocess2 1154 1150 929 pts/0 00:00:00 subprocess2 1155 1150 929 pts/0 00:00:00 subprocess2 1159 1159 929 pts/0 00:00:00 mainprocess 1160 1159 929 pts/0 00:00:00 mainprocess 1162 1159 929 pts/0 00:00:00 subprocess1 1163 1159 929 pts/0 00:00:00 subprocess1 1161 1159 929 pts/0 00:00:00 mainprocess 1164 1159 929 pts/0 00:00:00 subprocess2 1165 1159 929 pts/0 00:00:00 subprocess2 1166 1159 929 pts/0 00:00:00 subprocess2</pre>	

Question 7

Use `ps tree` command with option that show the number of threads to each process.

Command
<code>pstree -p -t</code>
Output
<pre> chew@secr2043:~\$ pstree -p -t systemd(1)---ModemManager(756)---{gdbus}(781) ---{gmain}(770) ---{pool-spawner}(777) -agetty(788) -cron(729) -dbus-daemon(639) -fwupd(984)---{GUsbEventThread}(987) ---{gdbus}(990) ---{gmain}(985) ---{libusb_event}(986) ---{pool-spawner}(988) -multipathd(358)---{multipathd}(382) ---{multipathd}(383) ---{multipathd}(384) ---{multipathd}(385) ---{multipathd}(386) ---{multipathd}(387) -polkitd(645)---{gdbus}(741) ---{gmain}(738) ---{pool-spawner}(739) -rsyslogd(684)---{in:imklog}(723) ---{in:imuxsock}(722) ---{rs:main Q:Reg}(725) -sshd(925)---sshd(926)---sshd(928)---bash(929)---mainprocess(1150)---mainprocess(1151)---subprocess1(1156) ---subprocess1(1157) ---mainprocess(1152)---subprocess2(1153) ---subprocess2(1154) ---subprocess2(1155) ---mainprocess(1159)---mainprocess(1160)---subprocess1(1162) ---subprocess1(1163) ---mainprocess(1161)---subprocess2(1164) ---subprocess2(1165) ---subprocess2(1166) ---pstree(1202) -systemd-journal(302) -systemd-logind(656) -systemd-network(513) -systemd-resolve(524) -systemd-timesyn(527)---{sd-resolve}(595) -systemd-udevd(365) -udisksd(658)---{cleanup}(769) ---{gdbus}(683) ---{gmain}(677) ---{pool-spawner}(680) ---{probing-thread}(757) -unattended-upgr(716)---{gmain}(807) </pre>

Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command
<code>sudo renice -n 10 -p 1156</code>
Output
<pre>[sudo] password for chew: 1156 (process ID) old priority 0, new priority 10</pre>

Question 9

Terminate all running processes with the name “mainprocess”.

Command
<code>pkill mainprocess</code>
Output
<pre>chew@secr2043:~\$ pkill mainprocess Main process (ID: 1150) received signal: 15. Terminating... Main process (ID: 1151) received signal: 15. Terminating... Main process (ID: 1161) received signal: 15. Terminating... Main process (ID: 1160) received signal: 15. Terminating... Main process (ID: 1159) received signal: 15. Terminating... [2]+ Done ./mainprocess Main process (ID: 1152) received signal: 15. Terminating...</pre>

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

C file name: fork_wait_example.c

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        // Child process
        printf("Child process: PID = %d, PPID = %d\n", getpid(), getppid());
    } else if (pid > 0) {
        // Parent process
        wait(NULL); // Wait for child to finish
        printf("Parent process: PID = %d\n", getpid());
    } else {
        // Fork failed
        printf("Fork failed.\n");
        return 1;
    }

    return 0;
}
```

Code typed in Linux:

```
chew@secr2043:~$ gcc fork_wait_example.c -o fork_wait_example
chew@secr2043:~$ ./fork_wait_example
```

Output:

```
chew@secr2043:~$ ./fork_wait_example
Child process: PID = 1248, PPID = 1247
Parent process: PID = 1247
```